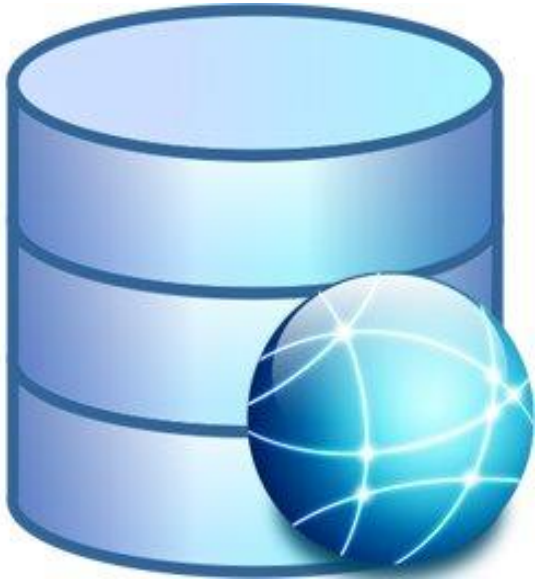
Database Systems

Lecture # 1

Noor ul Ain





Database Systems

- Online course content & coordination
  - Google Classroom

Recommended Text/ Reference Books



- Fundamentals of Database Systems,
  - *7th Edition by Elmasri & Navathe*
- Database System Concepts
  - *4th Edition by Abraham Silbershatz, Henery F. Korth, Sudarshan*
- An introduction to Database Systems
  - *By C.J. Date*

Grading Criteria (Marks Distribution)



- Tentative grading criteria is as follows:
  - Quizzes (10%)
  - Assignments (10%)
  - Project(10)%
  - Mid Term Exams (30%)
  - Final Exam (40%)



Grading Policy

- There is simply no chance of extension in any of the deadline, what so ever
- You can request for re-checking of any of your evaluation as per following rules;

  Exams: Same day

  Assignments: 2 days after handing-over

  Quizzes: 2 days after handing over

  Warning! After due time, request will not be considered even if it's genuine.

  Warning! Keep checking your FLEX regularly and don't come in the end with bulk of queries in hand which you never presented earlier and now you are on the Edge.



General Guidelines

- Cheating cases are intolerable. You will be given negative marks for cheated stuff irrespective of the fact that, you were provider or the other one. Your cheating in exam will make it easy for you to step down from the Course with an **'F' grade**. . . ☹
- Plagiarism in one item of the assessment instrument will result in cancellation of all items of corresponding instrument.
- There will be **NO RE-TAKE** of any Quiz, Assignment and Project.
- Quiz is inevitable so always, expect a One ☺ [at least, one in every week]



General Guidelines

- You have to depend on yourself to have a good grade in the course
  - *You need to appear in demo to have it graded*
  - *Grading will be individual even for group tasks. So better to shine with your own work in hand otherwise don't complaint*



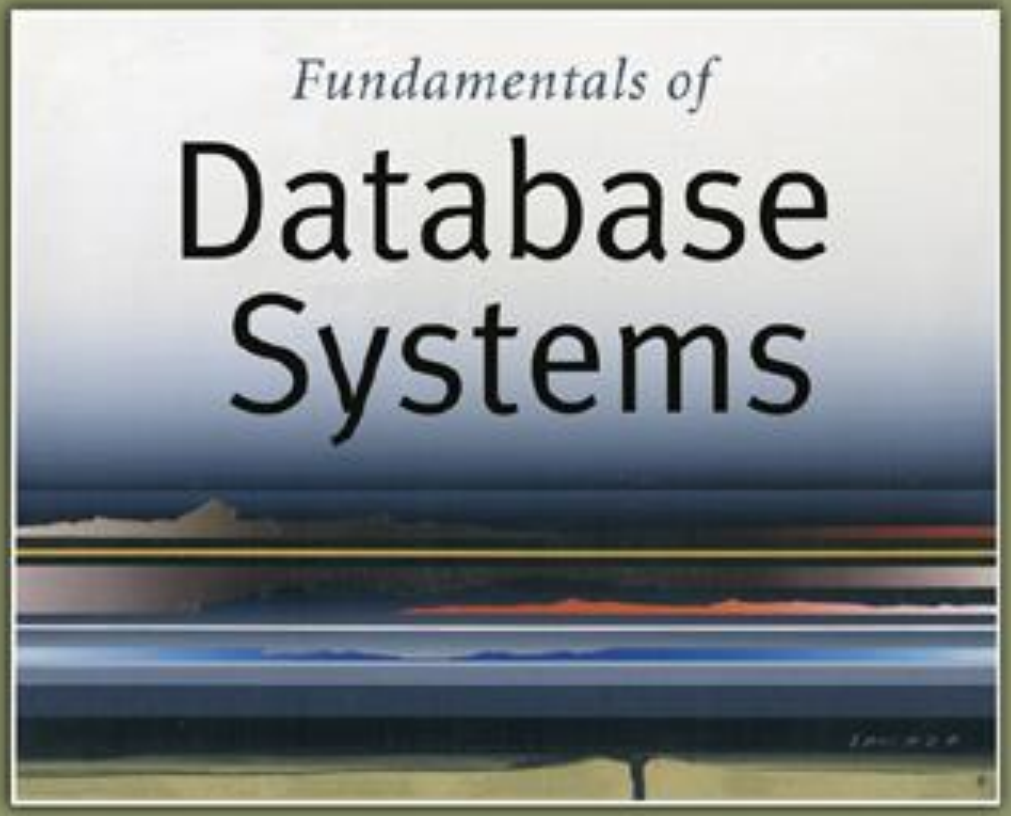
Course Contents

- Introduction to course, Introduction to Databases
- Database Systems Concepts & Architecture, Data Models
- Relational Model & Constraints
- SQL Programming
- Relational Algebra
- Data Modeling & Entity Relationships (ER)
- Enhanced Entity Relationships (EER)



Course Contents (Cont…)

- Database Design, Mapping from conceptual to logical model
- Functional Dependencies & Normalization
- Transaction Processing and concurrency control
- Introduction to Query Processing & Optimization
- Distributed Database Management System
- Data Warehouse and BI
- Big Data and NoSQL



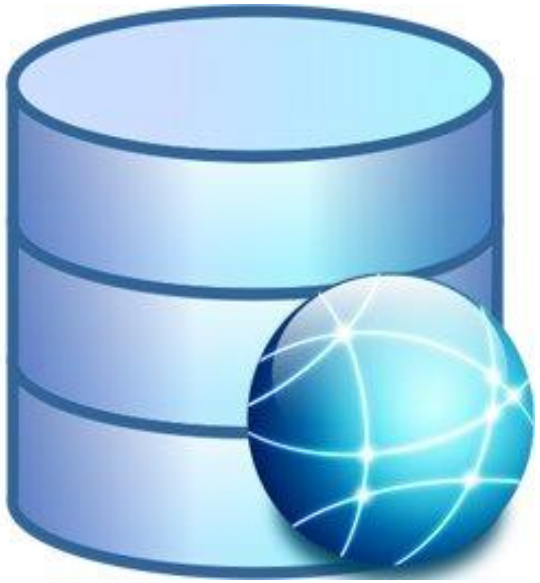
5th Edition

Elmasri / Navathe

Database Systems

Chapter 1

Introduction: Databases and Database Users



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Outline

- Types of Databases and Database Applications
- Basic Definitions
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach
- Database Users
- Advantages of Using the Database Approach
- When Not to Use Databases

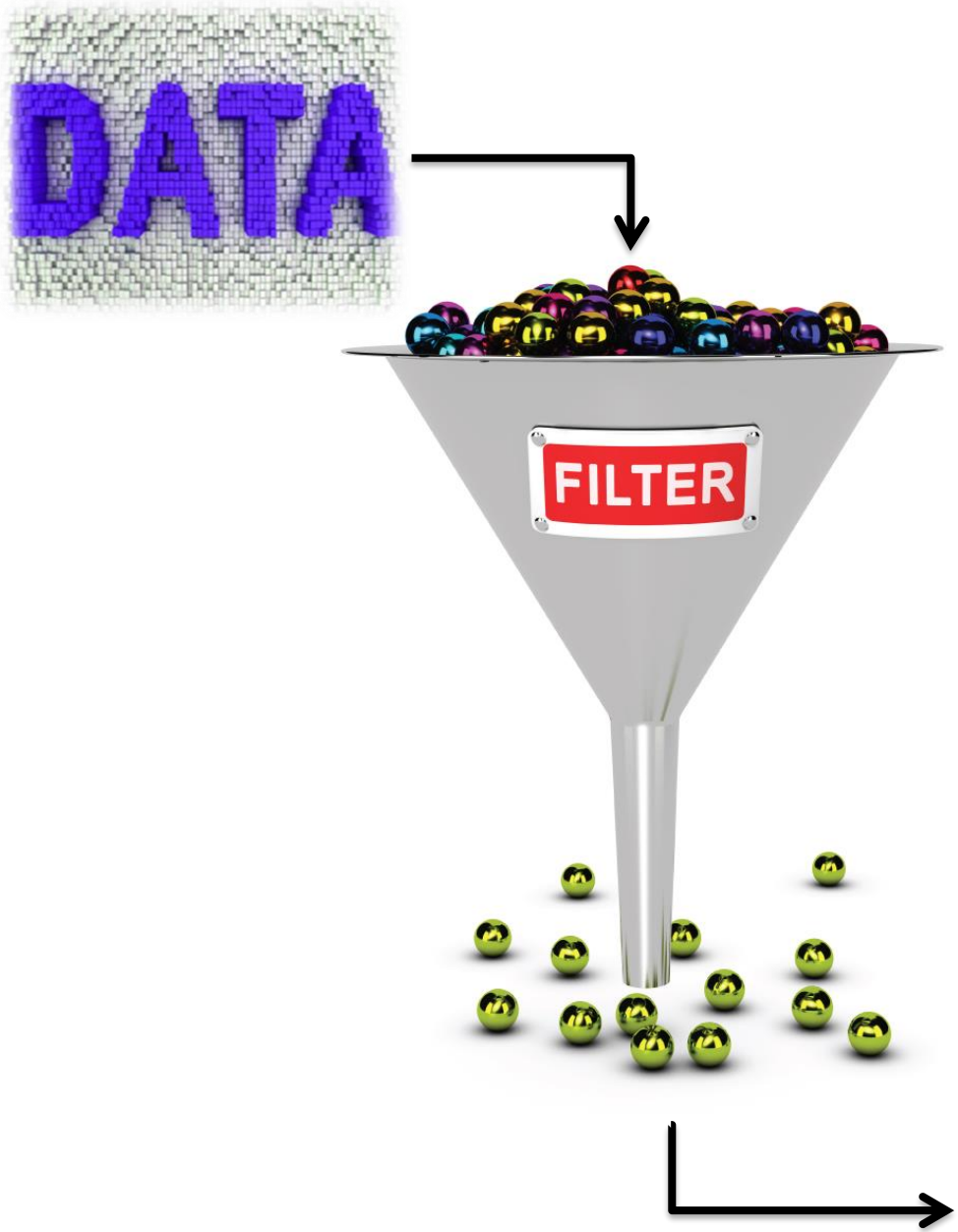
Types of Databases and Database Applications



- Traditional Applications:
  - Numeric and Textual Databases
- More Recent Applications:
  - Multimedia Databases
  - Geographic Information Systems (GIS)
  - Data Warehouses
  - Real-time and Active Databases
  - Many other applications
- First part of book focuses on traditional applications
- *A number of recent applications are described later in the book (for example, Chapters 24,26,28,29,30)*


DATA


RAW FACTS


DATA
FILTER


INFORMATION
KNOWLEDGE
LEARNING



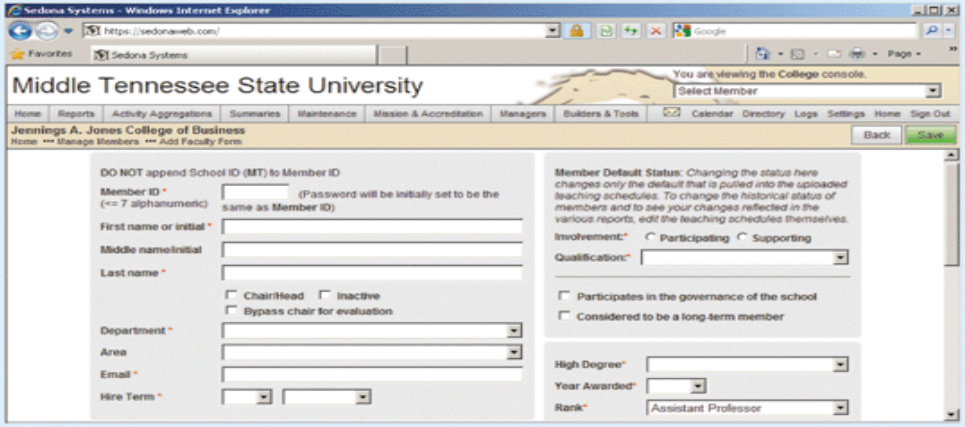
Data and Information …

- Data: is known recorded facts which has specific meanings or interpretation.
  - Example: the numbers 10.5, 22, and 119 are data
- Information: is a precise, understandable and specific representation of data.
  - Example: The temperature of room 119 in building 22 is 10.5, is an information
- Knowledge: is something which is derived or inferred from available information using some level of intelligence.
  - Example: Based on experience, the above information can be used to infer that the room is quite cold and could cause some inconvenience if we work on that room for a long period of time with out wearing worm clothes.

Data vs. Information …

FIGURE 1.1 Transforming raw data into information

a) Data entry screen



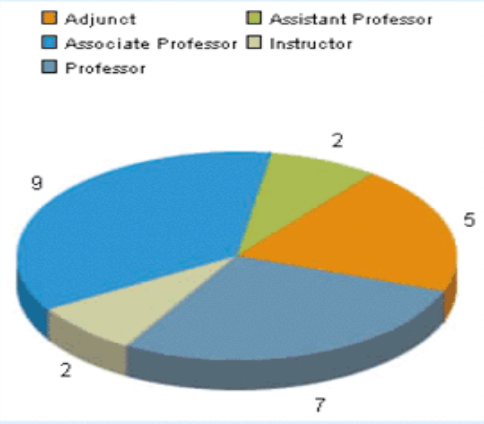
b) Raw data

Id	LastName	MidName	FirstName	DeptCode	Office	Email	Rank	HireYear	Degree
1	Washinghton	A.	George	MGMT	N135	gwashington@mtsu.edu	Professor	2001	Ph.D.
2	Adams		John	FIN	N313	jadams@mtsu.edu	Professor	1984	Ph.D.
3	Jefferson	L.	Thomas	ECON		tjefferson@mtsu.edu	Instructor	2002	M.B.A.
4	Madison	D.	James	FIN	N236	jmadison@mtsu.edu	Associate Professor	1994	Ph.D.
5	Monroe	N.	James	ACCT	N411	jmonroe@mtsu.edu	Assistant Professor	1995	Ph.D.
6	Adams	Q.	John	ACCT	N418	jqadams@mtsu.edu	Associate Professor	1989	Ph.D.
7	Jackson	C.	Andrew	ECON	N303	ajackson@mtsu.edu	Associate Professor	1999	Ph.D.
8	Van Buren	T.	Martin	FIN	N306	mvanburen@mtsu.edu	Professor	1988	Ph.D.
9	Harrison	R.	William	MKTG	N118	wharrison@mtsu.edu	Professor	1994	Ph.D.
10	Tyler	M.	John	MGMT		JTyler@mtsu.edu	Assistant Professor	2000	Ed.D.
11	Polk		Cheryl	MKTG	N143	cpolk@mtsu.edu	Associate Professor	2002	Ph.D.
12	Taylor	G.	Zachary	ACCT	N415	ztaylor@mtsu.edu	Associate Professor	1996	Ph.D.
13	Fillmore		Millard	JCB	N219	mfillmore@mtsu.edu	Professor	1992	Ph.D.
14	Pierce	A.	Franklin	MKTG	N359	pfranklin@mtsu.edu	Instructor	2005	M.B.A.
15	Buchanen	T.	James	MGMT	N146	jbuchanen@mtsu.edu	Associate Professor	1996	D.B.A.
16	Lincoln	W.	Larry	MGMT	N150	llincoln@mtsu.edu	Associate Professor	1996	Ph.D.
17	Johnson		Andrew	ISYS	N360	ajohnson@mtsu.edu	Professor	1987	Ph.D.
18	Grant		Katie	MKTG	N120	kgrant@mtsu.edu	Assistant Professor	1989	D.B.A.
19	Rutherford		Hayes	ACCT	N408	hrutherford@mtsu.edu	Professor	1992	Ph.D.
20	Garfield	T.	Denise	ACCT		dgarfield@mtsu.edu	Assistant Professor	2018	Ph.D.
21	Arthur		Emily	ACCT	N413	earthur@mtsu.edu	Associate Professor	2003	J.D.
22	Clevenland	G.	Robert	ACCT	N401	rcleveland@mtsu.edu	Associate Professor	1997	Ph.D.
23	Harrison	X.	Patricia	BULA	N406	pharrison@mtsu.edu	Associate Professor	2001	J.D.
24	McKinley	B.	Priscilla	ISYS	N363	pmckinley@mtsu.edu	Adjunct	1994	M.S.
25	Roosevelt	F.	Hillary	MGMT	N104	hroosevelt@mtsu.edu	Associate Professor	2002	Ph.D.
26	Wilson		Laura	BCEN	N448	lwilson@mtsu.edu	Professor	1992	Ph.D.
27	Harding		Warren	MKTG	N114	wharding@mtsu.edu	Professor	1984	Ed.D.
28	Coolidge		Calvin	ECON	N316	ccoolidge@mtsu.edu	Professor	1975	Ph.D.
29	Hoover		Lisa	MGMT		lhoover@mtsu.edu	Adjunct	1978	M.B.A.
30	Truman		Betty	ACCT	N416	btruman@mtsu.edu	Professor	1971	Ed.D.
31	Johnson		Robert	BCEN	N240	rjohnsonr@mtsu.edu	Professor	2001	Ph.D.

c) Information in summary format

Rank	COUNT	%/INFS	TOT/COL	%/COL. TOT.	%/COL. FAC.
Adjunct	5	20.00%	23	21.74%	3.27%
Assistant Professor	2	8.00%	28	7.14%	1.31%
Associate Professor	9	36.00%	37	24.32%	5.88%
Instructor	2	8.00%	18	11.11%	1.31%
Professor	7	28.00%	47	14.89%	4.58%

d) Information in graphical format



SOURCE: Course Technology/Cengage Learning
Data entry screen courtesy of Sedona Systems, 2011.
Information screens courtesy of JCBDashboard, 2011.



Basic Definitions

- **Database:**
  - A collection of related data.
- **Data:**
  - Known facts that can be recorded and have an implicit meaning.
- **Mini-world:**
  - Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.
- **Database Management System (DBMS):**
  - A software package/ system to facilitate the creation and maintenance of a computerized database.
- **Database System:**
  - The DBMS software together with the data itself. Sometimes, the applications are also included.

Simplified database system environment

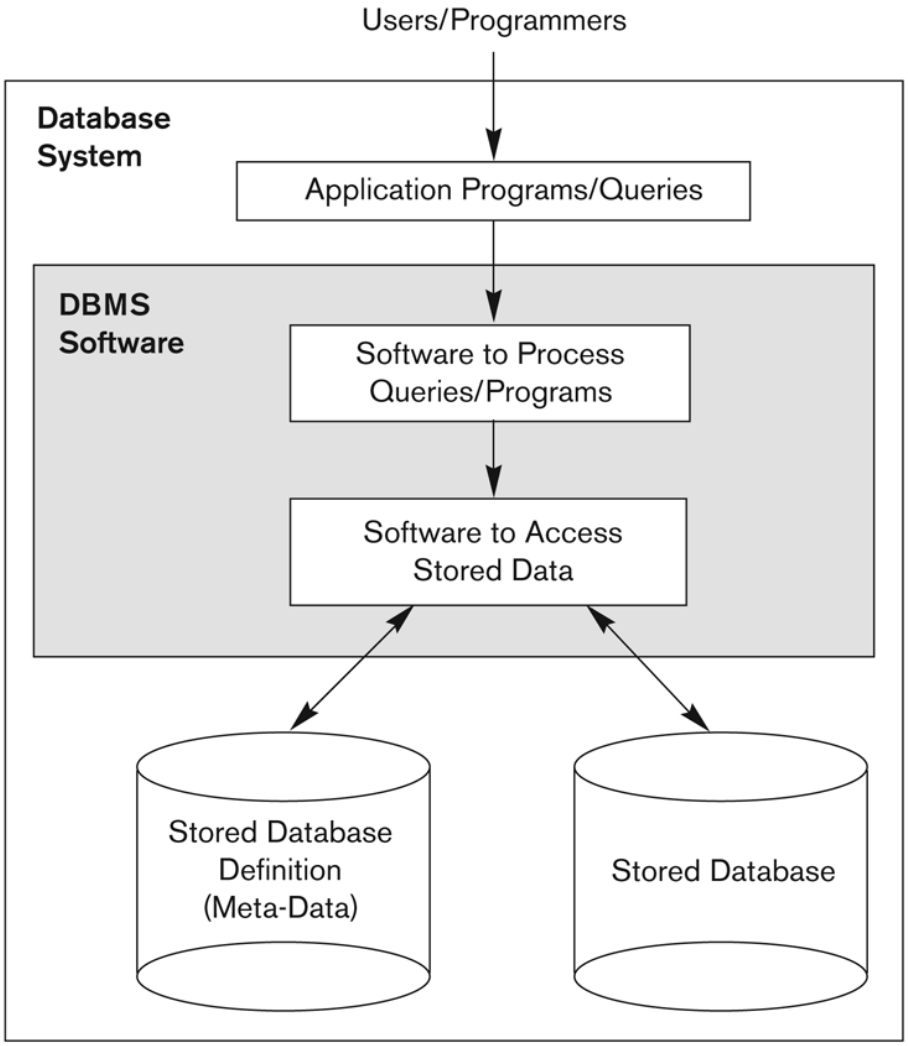


Figure 1.1
A simplified database system environment.



Data Base Management Systems

- A special software is accordingly needed to make the preceding tasks easier
- This software is known as Data Base Management System (DBMS)
- DBMSs provide automatic:
  - Data independence
  - Efficient data access
  - Data integrity and security
  - Data administration
  - Concurrent access and crash recovery
  - Reduced application development and tuning time



Typical DBMS Functionality

- *Define* a particular database in terms of its data types, structures, and constraints
- *Construct* or Load the initial database contents on a secondary storage medium
- *Manipulating* the database:
  - Retrieval: Querying, generating reports
  - Modification: Insertions, deletions and updates to its content
  - Accessing the database through Web applications
- *Processing* and *Sharing* by a set of concurrent users and application programs – yet, keeping all data valid and consistent



Typical DBMS Functionality

- Other features:
  - Protection or Security measures to prevent unauthorized access
  - “Active” processing to take internal actions on data
  - Presentation and Visualization of data
  - Maintaining the database and associated programs over the lifetime of the database application
    - Called database, software, and system maintenance

Database Applications



- Banking: all transactions
- Airlines: reservations, schedules
- Universities: registration, grades
- Sales: customers, products, purchases
- Manufacturing: production, inventory, orders, supply chain
- Human resources: employee records, salaries, tax deductions

Example of a Database (with a Conceptual Data Model)



- **Mini-world for the example:**
  - Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
  - STUDENTs
  - COURSEs
  - SECTIONs (of COURSEs)
  - (academic) DEPARTMENTs
  - INSTRUCTORs

Example of a Database (with a Conceptual Data Model)



- **Some mini-world *relationships*:**
  - SECTIONs *are of specific* COURSEs
  - STUDENTs *take* SECTIONs
  - COURSEs *have prerequisite* COURSEs
  - INSTRUCTORs *teach* SECTIONs
  - COURSEs *are offered by* DEPARTMENTs
  - STUDENTs *major in* DEPARTMENTs

- Note: The above entities and relationships are typically expressed in a conceptual data model, such as the ENTITY-RELATIONSHIP data model (see Chapters 3, 4)

Example of a simple database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2
A database that stores student and course information.



Learning Outcomes

- After finishing this course you will be able to:

1. Describe a wide range of data involved in real-world organizations using the entity-relationship (ER) data model

2. Explain how to translate an ER diagram into a relational database

3. Analyze and apply a formal query language, relational calculus and algebra

4. Indicate how SQL builds upon relational calculus and algebra and effectively apply SQL to create, query and manipulate relational databases

5. Design and develop multi-tiered, full-fledged standalone and web-based applications with back-end databases

6. Appreciate how DBMSs create, manipulate and manage files of fixed-length and variable-length records on disks

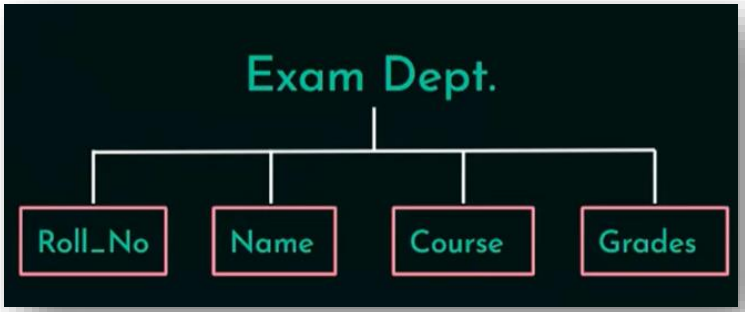


Drawbacks of File System
(File System Vs DBMS)

1. Data Redundancy & Data Inconsistency
2. Difficulty in Accessing Data
3. Data Isolation
4. Integrity Problems
5. Atomicity Problems
6. Concurrent Access Anomalies
7. Security Problems

File System

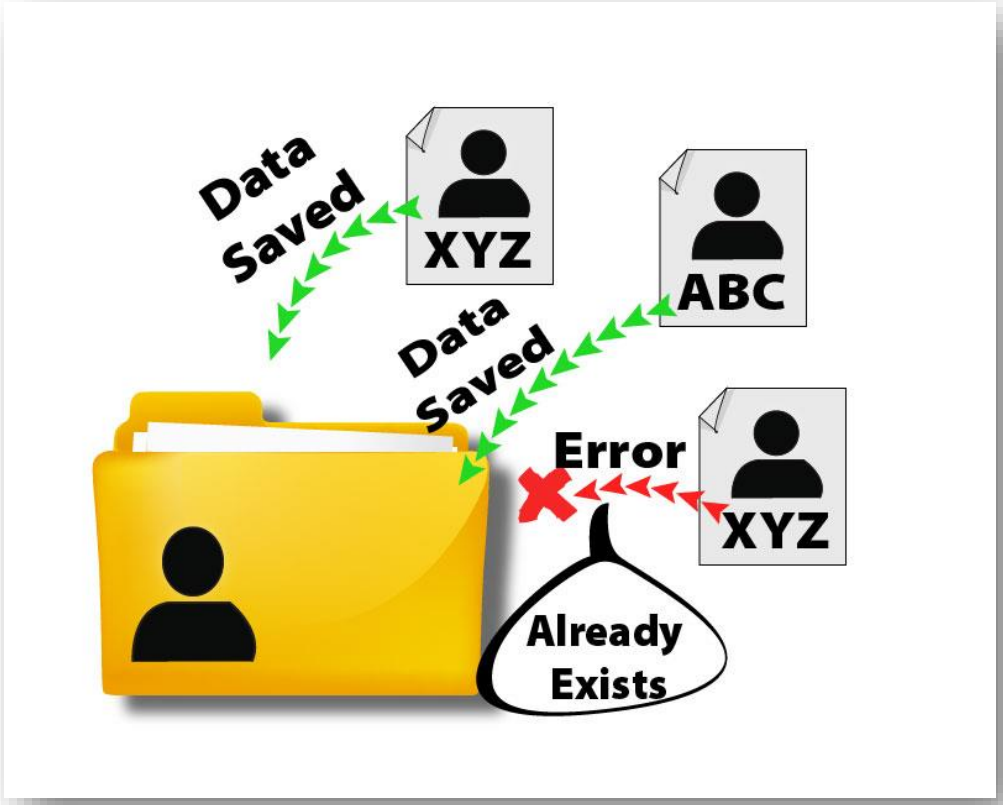




DBMS



1. Data Redundancy & Data Inconsistency





RegNo	S_Name	PERSONAL_NO	ADDRESS	FEE DUE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	5-10-10, 2/9 BRODIPET, GUNTUR	2000
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	1500
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	NIL

OFFICE



RegNo	S_Name	PERSONAL_NO	ADDRESS	BOOKS FINE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	5-10-10, 2/9 BRODIPET, GUNTUR	20
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	NIL
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	15

LIBRARY



RegNo	S_Name	PERSONAL_NO	ADDRESS	GRADE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	5-10-10, 2/9 BRODIPET, GUNTUR	B
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	A
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	C

Accounts



RegNo	S_Name	PERSONAL_NO	ADDRESS	FEE DUE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	5-10-10, 2/9 BRODIPET, GUNTUR	2000
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	1500
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	NIL

OFFICE



RegNo	S_Name	PERSONAL_NO	ADDRESS	BOOKS FINE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	5-10-10, 2/9 BRODIPET, GUNTUR	20
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	NIL
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	15

LIBRARY



RegNo	S_Name	PERSONAL_NO	ADDRESS	GRADE
14BQ1A0501	ACHYUTA NAGA BHAVA MANOJ	1234567890	15-11/B, 4/8 ARUNDALPET, GUNTUR	B
14BQ1A0502	ADDANKI KIRANMAI	2225557779	3-11-12/24, SANTOSH NAGAR, GUNTUR	A
14BQ1A0503	ALLA SIVA NAGA DIVYA SREE	1238901123	4-11/A, VIDHYANAGAR, TENALI	C

Accounts

2. Difficulty in Accessing Data



OFFICE



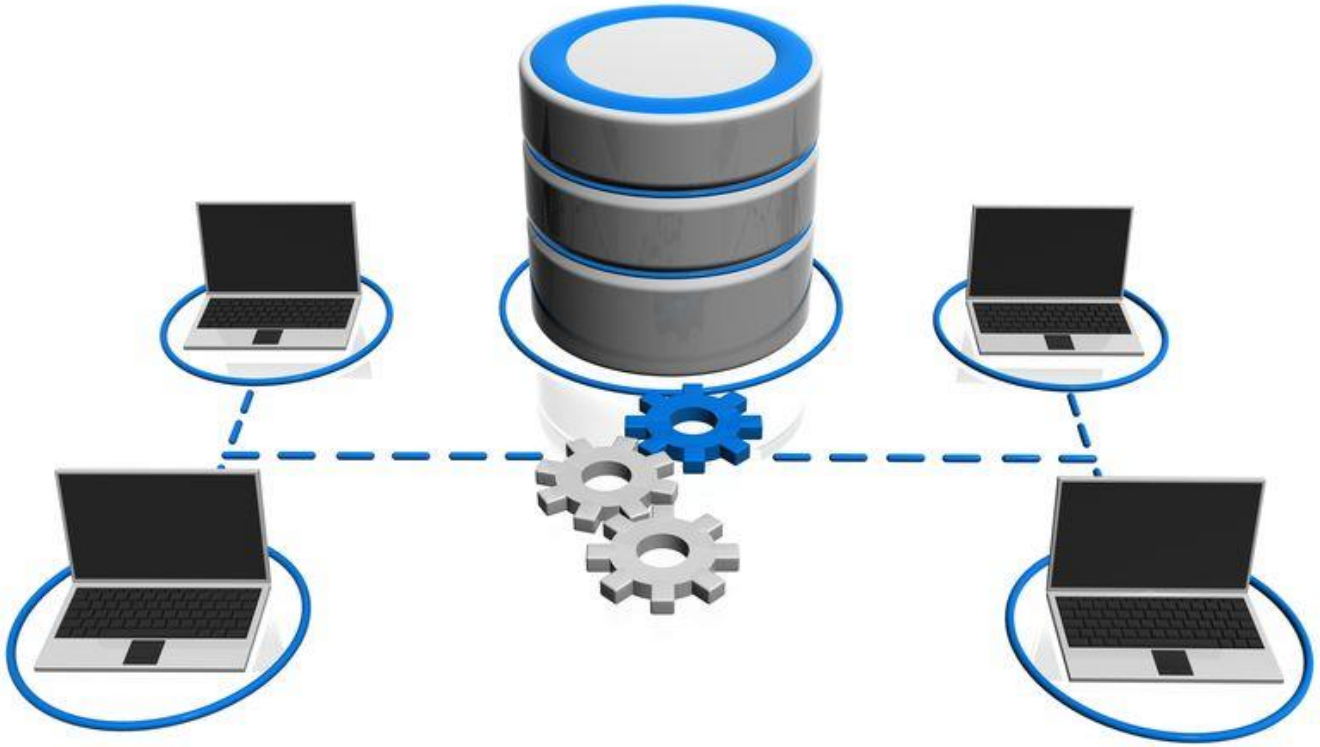


LIBRARY

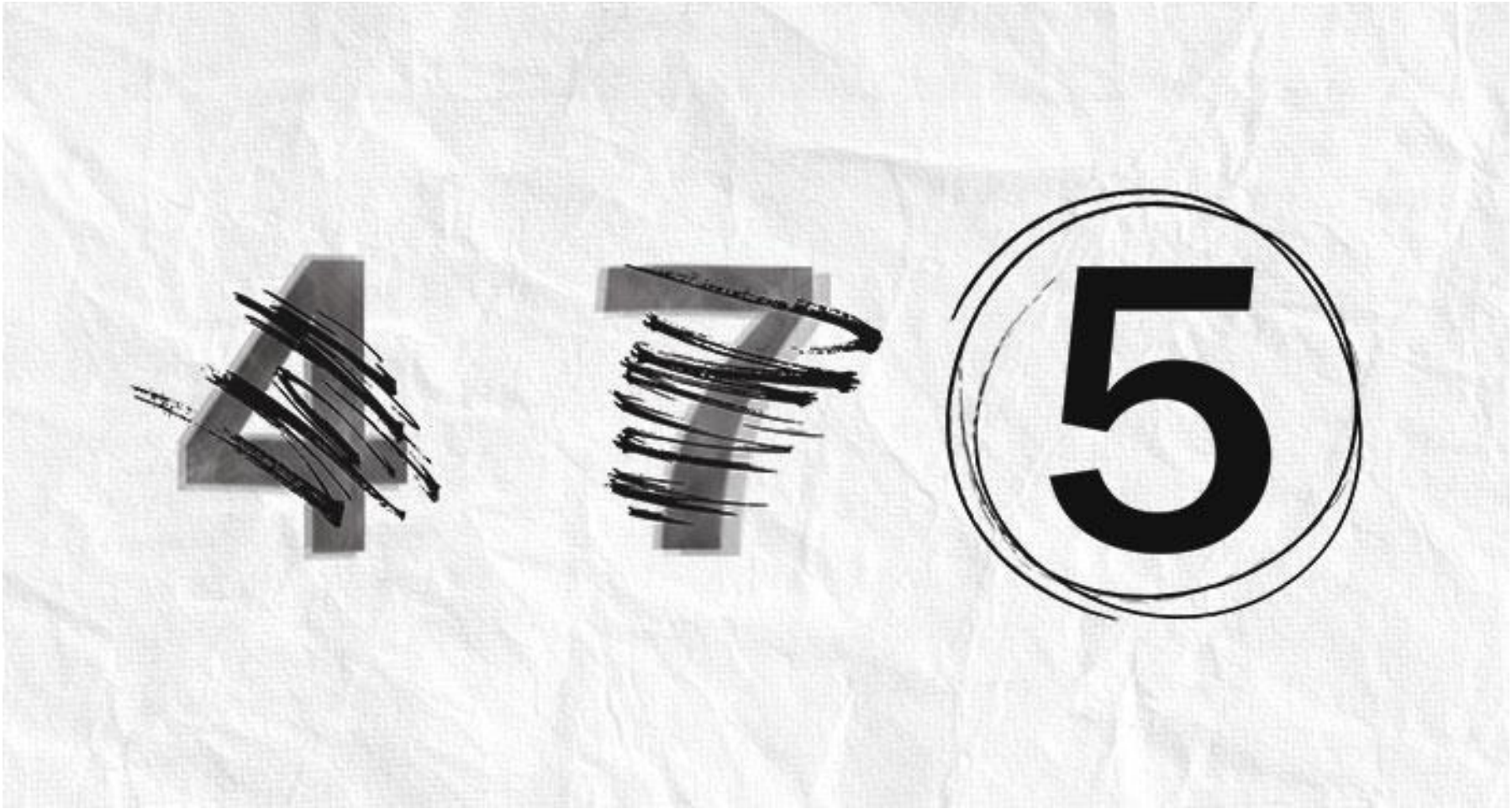


ACCOUNT

3. Data Isolation



4. Integrity Problems



5. Atomicity Problems



Account A

Account B

Initial Balance: 1000/-

Initial Balance: 2000/-

Fund Transfer of 800/- from Account A to Account B

6. Concurrent Access Anomalies



7. Security





Purpose of DBMS

- Database systems offer solutions to all the above problems

Advantages of DBMS over file system



1. Fast Data Access: The data response time increases in DBMS.

2. Minimized Data Redundancy: DBMS has different constraints using them same data can't be stored in more than one places.

3. Data Consistency: Since DBMS solves the problem of data redundancy, the problem of data consistency is automatically solved.

Advantages of DBMS over file system



4. No attributes for accessing the data: Here, we don't need to know the location of the file. The user makes a request from any web application or app and the server responds accordingly.

5. Concurrent Access: Multiple users can access the database at the same time when we are using the Database Management System.

6. Security: We have role-based access control in DBMS. Each user has a different set of access thus the data is secured from problems like data leaks, misuse of data etc.



A Motivating Scenario

- Pakistan Foundation (PF) has a “large” collection of data (say 500GB) on employees, students, universities, research centers, etc.,
- This data is accessed frequently by many people
- Queries on data must be answered quickly
- Changes made to the data by different people should be applied *consistently*
- Access to certain parts of data (e.g., salaries) must be *restricted*
- This data should survive system crashes and failures

Performance (Concurrency Control)

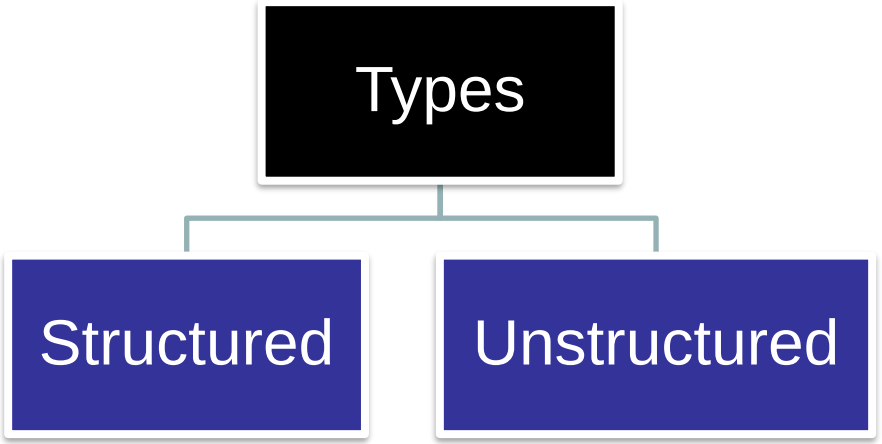
Performance (Response Time)

Correctness (Consistency)

Correctness (Security)

Correctness (Durability and Atomicity)

Types of Data

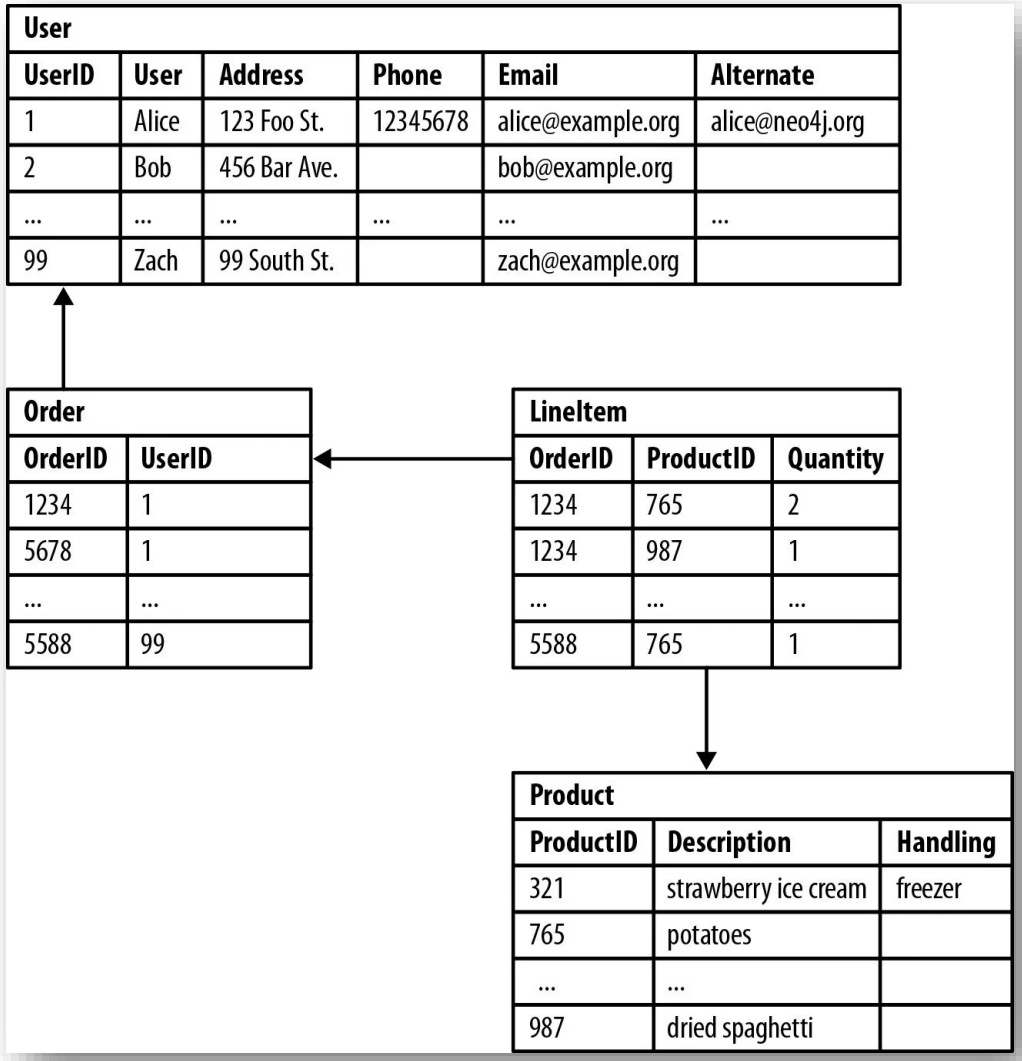




Structured Data

- Stored in tabular format
- Clearly defined
- Data is stored in a pre-defined data model
- Think of data that fits neatly within fixed fields and columns in relational databases and spreadsheets.
- Examples of structured data include
  - names, dates, addresses,
  - credit card numbers,
  - stock information,
  - geolocation, and more.

Structured Data



User

UserID	User	Address	Phone	Email	Alternate
1	Alice	123 Foo St.	12345678	alice@example.org	alice@neo4j.org
2	Bob	456 Bar Ave.		bob@example.org	
...	...	...	...	...	...
99	Zach	99 South St.		zach@example.org	

Order

OrderID	UserID
1234	1
5678	1
...	...
5588	99

LineItem

OrderID	ProductID	Quantity
1234	765	2
1234	987	1
...	...	...
5588	765	1

Product

ProductID	Description	Handling
321	strawberry ice cream	freezer
765	potatoes	
...	...	
987	dried spaghetti	

RDBMS

Structured data is stored in relational databases

Unstructured Data









- No predefined structure
- No data model
- Irregular and ambiguous
- Examples of unstructured data include
  - text,
  - video files,
  - audio files,
  - mobile activity,
  - social media posts,
  - satellite imagery,
  - surveillance imagery
- Non-relational or NoSQL databases are the best fit for managing unstructured data.

Types of Databases

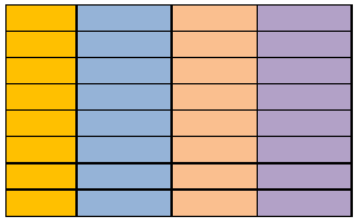
- SQL DB are structured DBs (DBMS)
- NoSQL database is
  - a next-generation DB
  - used for big data, mainly retrieval purpose
  - horizontally scalable
  - Apache Cassandra, Apachi HBase, Google Bigtable and MongoDB
- Graph, Document, Key-value and Column Database are also known as NoSQL
- Cloud DBs MS Azure & Amazon AWS



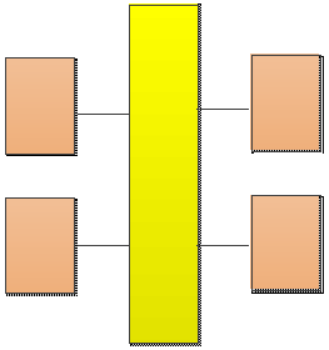
Architectures

SQL Databases

Relational

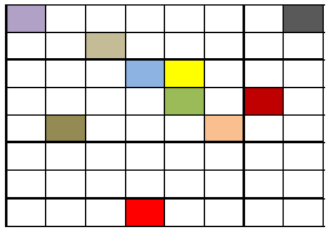


Analytical (OLAP)

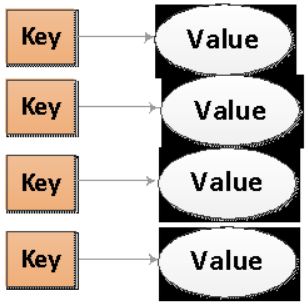


NoSQL Databases

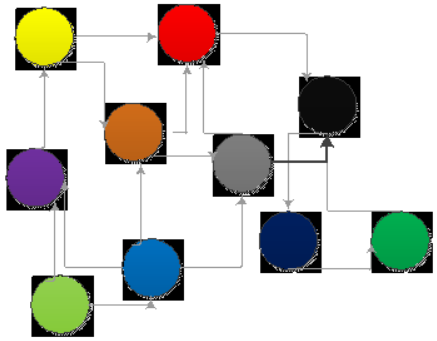
Column Family



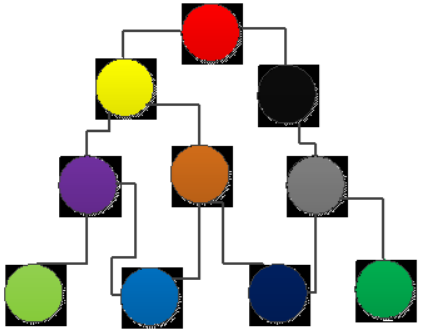
Key-Value



Graph



Document



Examples DBMS

- Oracle
- IBM DB2
- Ingress
- Teradata
- MS SQL Server
- MS Access
- MySQL





Database Design

- Refers to the activities that focus on the design of the database structure that will be used to store and manage end-user data.
- A database that meets all user requirements does not just happen; its structure must be designed carefully.
- Even a good DBMS will perform poorly with a badly designed database.

Database Design

- Consider the data shown below which illustrates the efforts of an organization to keep records about its employees and their skills.

FIGURE 1.4 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?

Do we add another column?

ID	ENum	Name	Title	HireDate	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
1	02345	Brian Oates	DBA	2/14/1995	Basic Database Management	2/14/2002	Advanced Database Management	2/14/2005	Basic Web Design	8/9/2003
2	08273	Marco Bienz	Analyst	7/28/2006	Basic Web Design	3/8/2009	Advance Process Modeling	8/19/2012		
3	06234	Jasmine Patel	Programmer	8/10/2005	Basic Web Design	8/10/2007	Advanced C# programming	8/10/2007	Basic DB manipulation	1/29/2012
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2002	Advanced Spreadsheets	6/20/2011				
5	13567	Almond, Robert	Analyst	9/30/2012	Basic Process Modeling	9/30/2014	Basic Database Design	5/23/2015		
6	10282	Richardson, Amanda	Clerk	4/11/2011						
7	09382	Susan Mathis	Database Programmer	8/2/2010	Basic DB Design	8/2/2012	Basic Database Manipulation	8/2/2012	Advanced DB Manipulation	5/1/2013
8	14311	Duong, Lee	Programmer	9/1/2014	Basic Web Design	9/1/2016				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2010	Advanced Spreadsheets	5/16/2013	Basic Web Design	5/16/2013		
12	13383	Raymond F. Matthews	Programmer	3/12/2012	Basic C# Programming	3/12/2014				
13	09283	Chavez, Juan	Clerk	7/4/2010						
14	04893	Patricia Richards	DBA	6/11/2004	Advanced Database Management	6/11/2006	Advanced Database Manipulation	9/20/2012		
15	13932	Lee, Megan	Programmer	9/23/2013						

Database Design

1. Difficult to produce an alphabetical listing of employees based on their last names
2. How to efficiently count how many possess which skill-set
   - Issue due to Skill 1 and Skill 2
3. Different names for same skill
4. How to add more columns, if an employee acquires another skill

FIGURE 1.4 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?

Do we add another column?

ID	ENum	Name	Title	HireDate	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
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3	06234	Jasmine Patel	Programmer	8/10/2005	Basic Web Design	8/10/2007	Advanced C# programming	8/10/2007	Basic DB manipulation	1/23/2012
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2002	Advanced Spreadsheets	6/20/2011				
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6	10282	Richardson, Amanda	Clerk	4/11/2011						
7	09382	Susan Mathis	Database Programmer	8/2/2010	Basic DB Design	8/2/2012	Basic Database Manipulation	8/2/2012	Advanced DB Manipulation	5/1/2013
8	14311	Duong, Lee	Programmer	9/1/2014	Basic Web Design	9/1/2016				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2010	Advanced Spreadsheets	5/16/2013	Basic Web Design	5/16/2013		
12	13383	Raymond F. Matthews	Programmer	3/12/2012	Basic C# Programming	3/12/2014				

Better Database Design



Table name: EMPLOYEE

Employee_ID	Employee_FName	Employee_LName	Employee_HireDate	Employee_Title
02345	Johnny	Jones	2/14/1995	DBA
03373	Franklin	Johnson	3/15/2002	Purchasing Agent
04893	Patricia	Richards	6/11/2004	DBA
06234	Jasmine	Patel	8/10/2005	Programmer
08273	Marco	Bienz	7/28/2006	Analyst
09002	Ben	Joiner	5/20/2010	Clerk
09283	Juan	Chavez	7/4/2010	Clerk
09382	Jessica	Johnson	8/2/2010	Database Programmer
10282	Amanda	Richardson	4/11/2011	Clerk
13383	Raymond	Matthews	3/12/2012	Programmer
13567	Robert	Almond	9/30/2012	Analyst
13932	Megan	Lee	9/29/2013	Programmer
14311	Lee	Duong	9/1/2014	Programmer

Table name: CERTIFIED

Employee_ID	Skill_ID	Certified_Date
02345	100	2/14/2002
02345	110	8/9/2003
02345	180	2/14/2006
03373	120	6/20/2011
04893	180	6/11/2006
04893	220	9/20/2012
06234	110	8/10/2007
06234	200	8/10/2007
06234	210	1/29/2012
08273	110	3/8/2009
08273	190	8/19/2012
09002	110	5/16/2013
09002	120	5/16/2013
09382	140	8/2/2012
09382	210	8/2/2012
09382	220	5/1/2013
13383	170	3/12/2014
13567	130	9/30/2014
13567	140	5/23/2015
14311	110	9/1/2016

Table name: SKILL

Skill_ID	Skill_Name	Skill_Description
100	Basic Database Management	Create and manage database user accounts.
110	Basic Web Design	Create and maintain HTML and CSS documents.
120	Advanced Spreadsheets	Use of advanced functions, user-defined functions, and macroing.
130	Basic Process Modeling	Create core business process models using standard libraries.
140	Basic Database Design	Create simple data models.
150	Master Database Programming	Create integrated trigger and procedure packages for a distributed environment.
160	Basic Spreadsheets	Create single tab worksheets with basic formulas
170	Basic C# Programming	Create single-tier data aware modules.
180	Advanced Database Management	Manage Database Server Clusters.
190	Advance Process Modeling	Evaluate and Redesign cross-functional internal and external business processes.
200	Advanced C# Programming	Create multi-tier applications using multi-threading
210	Basic Database Manipulation	Create simple data retrieval and manipulation statements in SQL.
220	Advanced Database Manipulation	Use of advanced data manipulation methods for multi-table inserts, set operations, and correlated subqueries.

Database design – another example

- Data Redundancy & Design

EmpNum	EmpName	DeptNum	DeptName
100	Jones	10	Accounting
150	Lau	20	Marketing
200	McCauley	10	Accounting
300	Griffin	10	Accounting

(a) One-Table Design

DeptNum	DeptName
10	Accounting
20	Marketing

EmpNum	EmpName	DeptNum
100	Jones	10
150	Lau	20
200	McCauley	10
300	Griffin	10

(b) Two-Table Design

Main Characteristics of the Database Approach



- **Self-describing nature of a database system:**
  - A DBMS **catalog** stores the description of a particular database (e.g. data structures, types, and constraints)
  - The description is called **meta-data**.
  - This allows the DBMS software to work with different database applications.
- **Insulation between programs and data:**
  - Called **program-data independence**.
  - Allows changing data structures and storage organization without having to change the DBMS access programs.

Example of a simplified database catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
….	….	…..
….	….	…..
….	….	…..
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerared type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

Figure 1.3
An example of a database catalog for the database in Figure 1.2.

Main Characteristics of the Database Approach (continued)



- **Data Abstraction:**
  - A **data model** is used to hide storage details and present the users with a conceptual view of the database.
  - Programs refer to the data model constructs rather than data storage details
- **Support of multiple views of the data:**
  - Each user may see a different view of the database, which describes **only** the data of interest to that user.

Main Characteristics of the Database Approach (continued)



- **Sharing of data and multi-user transaction processing:**
  - Allowing a set of **concurrent users** to retrieve from and to update the database.
  - *Concurrency control* within the DBMS guarantees that each **transaction** is correctly executed or aborted
  - *Recovery* subsystem ensures each completed transaction has its effect permanently recorded in the database
  - **OLTP** (Online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second.

Database Users

- Users may be divided into
  - Those who actually use and control the database content, and those who design, develop and maintain database applications (called "Actors on the Scene"), and
  - Those who design and develop the DBMS software and related tools, and the computer systems operators (called "Workers Behind the Scene").



Database Users

- Actors on the scene
  - **Database administrators:**
    - Responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring efficiency of operations.
  - **Database Designers:**
    - Responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.



Categories of End-users

- Actors on the scene (continued)
  - **End-users:** They use the data for queries, reports and some of them update the database content. End-users can be categorized into:
    - **Casual**: access database occasionally when needed
    - **Naïve** or Parametric: they make up a large section of the end-user population.
      - They use previously well-defined functions in the form of “canned transactions” against the database.
      - Examples are bank-tellers or reservation clerks who do this activity for an entire shift of operations.

Categories of End-users (continued)



- **Sophisticated:**
  - These include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities.
  - Many use tools in the form of software packages that work closely with the stored database.
- **Stand-alone:**
  - Mostly maintain personal databases using ready-to-use packaged applications.
  - An example is a tax program user that creates its own internal database.
  - Another example is a user that maintains an address book

Advantages of Using the Database Approach



- Controlling redundancy in data storage and in development and maintenance efforts.
  - Sharing of data among multiple users.
- Restricting unauthorized access to data.
- Providing persistent storage for program Objects
  - In Object-oriented DBMSs – see Chapters 20-22
- Providing Storage Structures (e.g. indexes) for efficient Query Processing

Redundancy

Figure 1.6
Redundant storage of Student_name and Course_name in GRADE_REPORT. (a) Consistent data. (b) Inconsistent record.

(a)

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Smith	112	MATH2410	B
17	Smith	119	CS1310	C
8	Brown	85	MATH2410	A
8	Brown	92	CS1310	A
8	Brown	102	CS3320	B
8	Brown	135	CS3380	A

(b)

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Brown	112	MATH2410	B

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

Advantages of Using the Database Approach (continued)



- Providing backup and recovery services.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.
- Enforcing integrity constraints on the database.
- Drawing inferences and actions from the stored data using deductive and active rules

Additional Implications of Using the Database Approach



- Potential for enforcing standards:
  - This is very crucial for the success of database applications in large organizations. **Standards** refer to data item names, display formats, screens, report structures, meta-data (description of data), Web page layouts, etc.
- Reduced application development time:
  - Incremental time to add each new application is reduced.

Additional Implications of Using the Database Approach (continued)

- Flexibility to change data structures:
  - Database structure may evolve as new requirements are defined.
- Availability of current information:
  - Extremely important for on-line transaction systems such as airline, hotel, car reservations.
- Economies of scale:
  - Wasteful overlap of resources and personnel can be avoided by consolidating data and applications across departments.

Historical Development of Database Technology



- Early Database Applications:
  - The Hierarchical and Network Models were introduced in mid 1960s and dominated during the seventies.
  - A bulk of the worldwide database processing still occurs using these models, particularly, the hierarchical model.
- Relational Model based Systems:
  - Relational model was originally introduced in 1970, was heavily researched and experimented within IBM Research and several universities.
  - Relational DBMS Products emerged in the early 1980s.

Historical Development of Database Technology (continued)



- Object-oriented and emerging applications:
  - Object-Oriented Database Management Systems (OODBMSs) were introduced in late 1980s and early 1990s to cater to the need of complex data processing in CAD and other applications.
    - Their use has not taken off much.
  - Many relational DBMSs have incorporated object database concepts, leading to a new category called *object-relationa*l DBMSs (ORDBMSs)
  - *Extended relational* systems add further capabilities (e.g. for multimedia data, XML, and other data types)

Historical Development of Database Technology (continued)



- Data on the Web and E-commerce Applications:
  - Web contains data in HTML (Hypertext markup language) with links among pages.
  - This has given rise to a new set of applications and E-commerce is using new standards like XML (eXtended Markup Language). (see Ch. 27).
  - Script programming languages such as PHP and JavaScript allow generation of dynamic Web pages that are partially generated from a database (see Ch. 26).
    - Also allow database updates through Web pages

Extending Database Capabilities



- New functionality is being added to DBMSs in the following areas:
  - Scientific Applications
  - XML (eXtensible Markup Language)
  - Image Storage and Management
  - Audio and Video Data Management
  - Data Warehousing and Data Mining
  - Spatial Data Management
  - Time Series and Historical Data Management

- The above gives rise to *new research and development* in incorporating new data types, complex data structures, new operations and storage and indexing schemes in database systems.

Emergence of Big Data Storage Systems and NOSQL Databases



- In the first decade of the twenty-first century, the proliferation of applications and platforms such as
  - social media Web sites,
  - large e-commerce companies,
  - Web search indexes, and
  - cloud storage/backup

  led to a surge in the amount of data stored on large databases and massive servers.
- New types of database systems were necessary to manage these huge databases—systems that would provide fast search and retrieval as well as reliable and safe storage of nontraditional types of data, such as social media posts and tweets.

Emergence of Big Data Storage Systems and NOSQL Databases



- Some of the requirements of these new systems were not compatible with SQL relational DBMSs (SQL is the standard data model and language for relational databases).
- The term *NOSQL* is generally interpreted as Not Only SQL, meaning that in systems that manage large amounts of data, some of the data is stored using SQL systems, whereas other data would be stored using NOSQL, depending on the application requirements.



When not to use a DBMS

- Main inhibitors (costs) of using a DBMS:
  - High initial investment and possible need for additional hardware.
  - Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- When a DBMS may be unnecessary:
  - If the database and applications are simple, well defined, and not expected to change.
  - If there are stringent real-time requirements that may not be met because of DBMS overhead.
  - If access to data by multiple users is not required.



When not to use a DBMS

- When no DBMS may suffice:
  - If the database system is not able to handle the complexity of data because of modeling limitations
  - If the database users need special operations not supported by the DBMS.



Summary

- Types of Databases and Database Applications
- Basic Definitions
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach
- Database Users
- Advantages of Using the Database Approach
- When Not to Use Databases

Reference

- Chapter # 1
- **Fundamentals of Database Systems,** by Ramez Elmasri and Shamkant Navathe, Addison Wesley, 7th Edition.